

RAGHAV PRATAP SINGH

SOFTWARE DEVELOPER

I am a B.Tech Computer Science Engineering student at SKIT Jaipur with a strong passion for software development, machine learning, and system optimization. Skilled in Python, C++, and Java, with experience in embedded systems. I have developed automation tools like MailLoop and LiNEDATA, built a battery safety monitoring model using ML, and created a prototype warranty system. I am interested in automation, AI-driven solutions, and optimizing system efficiency. Seeking opportunities to apply my technical expertise in a dynamic environment.

EDUCATION

- SKIT M&G, Jaipur**
B.Tech in Computer Science and Engineering (Pursuing)
2022 - 2026
- Kendriya Vidyalaya No. 5, Jaipur**
Senior Secondary Education
2021
- Kendriya Vidyalaya No. 5, Jaipur**
Secondary Education
2019

SKILLS

- Programming Languages:** Python, C, C++, Java
- Modules:** OpenCV, Selenium, NumPy, pandas, Matplotlib, mysql.connector, Tkinter, PyZBar, and csv
- Databases:** MySQL
- Arduino Development**
- Technical Documentation**

STRENGTHS

- Leadership
- Communication Skills
- Problem-Solving Skills
- Project Management
- Eager to Learn

CERTIFICATIONS

- Introduction to programming in C
- Programming in Modern C++
- The Joy of Computing using Python
- Programming In Java
- Fundamentals of Blockchain & Data Security
- Applications of AI & ML in IoT
- Soft Skills

PROJECTS

- Battery Management System**
 - Analyzes battery conditions to prevent EV explosions by isolating affected clusters.
 - It includes ML-driven automated battery switching technique for seamless experience.
- MailLoop**
 - Created MailLoop, a Python app using Tkinter for email tasks.
 - Features: user login, composing emails, selecting recipients from Excel, and sending with attachments.
 - Implemented automatic attachment handling by retrieving file names from a designated column in Excel.
 - Designed user-friendly interface for smooth navigation.
- LiNEDATA**
 - Developed a Python application using OpenCV and PyZBar for recording unpacking videos of return packages from online orders.
 - Implemented features to organize videos and images in designated folders with proper package IDs.
 - Streamlined the return filing process by facilitating efficient management of recorded media.
- Meesho Payment Scraper**
 - Developed a Selenium bot to automate data extraction from the Meesho online selling platform.
 - Scraped payment data for a selected month, including order details like sub-order number, SKU, status, and net order amount.
 - Categorized payment data into separate CSV files based on order status (e.g., returns, RTO, delivered orders, exchanges).
 - Handled error scenarios, such as element not found exceptions, to ensure smooth execution.
 - Improved efficiency by eliminating manual data entry, enabling streamlined analysis of sales and financial transactions.
- Gesture Controlled Robotic Arm**
 - Developed a Python script leveraging OpenCV and Mediapipe to detect arm movement gestures using a webcam. Designed to control the robotic arm through hand gestures captured by the webcam in real-time.
 - Utilized the HandTrackingModule from the cvzone library to track hand landmarks and estimate elbow angle.
 - Established serial communication with an Arduino microcontroller to control robotic arm motors based on transmitted angle data.
- Bluetooth Controlled Pick & Place Bot**
 - Developed a pick-and-place robot using Arduino for motor control and Bluetooth interfacing.
 - Developed a smartphone application using MIT App Inventor for remote control via Bluetooth.
- TRACK'EM**
 - Created an Automatic Number Plate Recognition (ANPR) system by harnessing the YOLOv8 object detection algorithm, ensuring precise and efficient vehicle tracking capabilities.
 - Integrated an easyOCR module for text extraction from license plates.
 - Implemented criminal vehicle detection feature within the system.

EXPERIENCE

- Python developer intern at Kistechnosoftware Pvt. Ltd.**
- C++ Summer Training**

EXTRA CURRICULAR

- Student Coordinator, Design Committee of the Smart India Hackathon 2023**
- Joint winner, Project Competition in annual techno-cultural fest - PRAVAH'24**
- Second Position, Digital Logo Making in annual techno-cultural fest - PRAVAH'23**
- Volunteering as a graphic designer for workshops conducted in college**
- Volunteering as an on-site support and coordination for workshops**